

PEA CORE - Primitive Ethical Axioms

v1.0 Public Ethical Review Specification

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The core does not decide what the world is, who holds final authority, or which human value must win. It preserves the conditions under which those questions may be asked, answered, contested, refused, reopened, and carried forward without silent erasure.

Canonicalization notice. PEA Core v1.0 is the public compact specification. PEAR v0.14.2 remains the reconstruction journal, stress-test history, and process record. Readers do not need PEAR to understand or apply this Core; it preserves the route by which the Core was assembled, challenged, and refined.

Abstract

PEA Core v1.0 is a compact, public, machine-addressable specification of the Primitive Ethical Axioms. It is not a universal moral proof, an autonomous decision-maker, or a substitute for affected people, law, professional duty, democratic institutions, or accountable judgment. It defines a bounded ethical review process for consequential actions under declared commitments, typed evidence, traceable authority, and contestable human governance.

The model receives a typed PECAN-valid or declared compatible protocol handoff, constructs a non-scalar ethical state, applies sixteen ethical positions and their derived corollaries, and returns a candidate normative judgment with reasons, conditions, unresolved remainder, contest routes, expiry, and reopening dependencies. Hard floors remain non-compensable: authority cannot replace consent, evidence cannot erase standing, and aggregate benefit cannot silently defeat privacy, remedy, or meaningful exit.

PEA occupies the normative review layer between a protocol-valid consequential crossing and an accountable human or institutional decision. PAL may supply structural references; PECAN determines whether a claim is being used to route consequence and whether the handoff is protocol-ready. PEA evaluates the proposed burden under declared ethical commitments. No step promotes description into permission, protocol validity into ethical approval, or ethical approval into execution authority.

1. Status, Role, and Authority Boundary

1.1 PEA Core versus PEA_r

PEA_r is the reconstruction branch: it contains the drafting history, full axiom arguments, amendments, failed formulations, tests, countermodels, and journal trace. PEA Core is the canonical compact model extracted from that work for implementation, review, publication, and integration.

Artifact	Primary role	Authority
PEA _r	Reconstruction journal, stress-test record, source detail	Historical and explanatory; not the runtime core
PEA Core	Canonical ethical model, schemas, guards, invariants, receipts	Controls the declared PEA evaluation and normative receipt within its adopted scope
PECAN	Consequential-crossing and authorization protocol	Controls protocol validity and handoff lineage

1.2 Candidate, not sovereign

PEA Core produces candidate normative judgments inside a declared PEA framework. It does not create the authority it records, execute the action it evaluates, appoint the final decision-maker, or declare its values mandatory for every society or domain.

Candidate judgment ≠ universal fact ≠ executable permission.

An institution may adopt PEA, reject it, fork it, or place another normative evaluator behind PECAN. Honest non-adoption is outside PEA; it is not approval by PEA. Once PEA is invoked for an action-version, withdrawal may stop further participation but may not erase findings, refusals, unresolved burdens, or receipt lineage already produced.

1.3 Public intent

PEA Core v1.0 is intended to be public, auditable, bounded, and non-authoritative. Public review is a strength: counterexamples, implementation failures, dependency gaps, hidden assumptions, and conflicting interpretations should remain visible. Public access does not convert the core into a universal regulator, and implementation success does not convert it into moral sovereignty.

2. Public Architectural Position

2.1 The public route

Layer	Question	May do	May not do
PAL	What center, boundary, trace, relation, and closure does the evidence support?	Describe typed structure and uncertainty	Derive PEA values or grant ethical permission
PECAN	Is a consequential crossing present, validly addressed, and protocol-ready?	Validate route, authority lineage, receipts, dependencies, and canonical identity	Return ethical PASS or REFUSE
PEA Core	How may the proposed crossing ethically proceed, pause, narrow, protect, repair, or refuse?	Apply ethical floors, conflict rules, and candidate judgment	Repair malformed PECAN lineage or rewrite PAL ontology
Accountable boundary / SeedPEA	Who decides, communicates, contests, and carries consequences?	Make governed decisions and human-readable releases	Claim the model made the decision or erase contest

2.2 No backward authority promotion

PAL description → PECAN routing → PEA evaluation → accountable decision.

Information may travel backward for correction or reopening, but authority does not. A PEA verdict cannot rewrite the PAL object that supplied its premise. A later PEA-governed act may create a new observable PAL event, but authorization alone does not make that event occur. PECAN approval cannot manufacture ethical approval, and PEA approval cannot repair a stale, replayed, malformed, or authority-promoted protocol packet.

2.3 Optional attachment

PEA is normatively independent of PAL. A PAL-like carrier can provide precise references to candidate centers, traces, attachments, and unresolved identity, but PEA may also be invoked by another sufficiently traceable system. Conversely, PAL may continue without PEA. The attachment is a bridge, not a derivation.

3. Core Input, State, and Output

3.1 Required protocol handoff

PEA Core evaluates only a typed handoff packet that is valid under the declared routing protocol. The public reference protocol for v1.0 is PECAN v1.0 or a declared compatible successor whose semantics, lineage, authority ceilings, and migration rules are explicit.

$H = (\text{proposal, crossing, route, fields, evidence, authority, receipts, dependencies, aperture, lineage}).$

At minimum, the packet preserves the proposal; consequential-crossing identity; routing state; applicable Prime-Ledger fields; evidence and authority ledgers; consent and standing references; dependency graph; unresolved remainder; boundary and aperture; version manifest; refusal lineage; and terminal receipt chain.

If the packet is malformed, stale, replayed, authority-promoted, semantically evasive, or not protocol-valid, PEA may return it, refuse it, or preserve it unresolved. It may not ethically bless a route whose protocol lineage is invalid.

3.2 Ethical-state record

$E = (C, S, K, A, V, P, R, T, G, Q).$

Symbol	Meaning
C	Affected-center and interaction structure
S	Standing and representation
K	Consent and consent-capacity
A	Authority, jurisdiction, scope, role, and grant lineage
V	Vulnerability, material risk, and burden role
P	Privacy, disclosure, persistence, and release depth
R	Reversibility, restoration, compensation, counteraction, and residual debt
T	Time horizon, agency-time distribution, delay, and expiry
G	Generational or successor carry
Q	Evidence, provenance, uncertainty, permitted claim layer, and unresolved remainder

The ethical state is not a scalar. Two packets with the same aggregate score may differ decisively because standing, consent, authority, privacy, remedy, or unresolved dependency is non-substitutable.

3.3 Candidate judgments

Judgment	Meaning
PASS	The proposed route satisfies the declared PEA requirements within scope.
PASS_WITH_CONDITIONS	The route may proceed only under explicit conditions, limits, review, or repair duties.
PROTECTIVE_ACTION	A separate minimally invasive protective crossing is justified; the original proposal is not promoted.
HOLD	The route is paused while a material dependency, contest, consent, authority, or safety issue is addressed.
UNRESOLVED	The required normative conclusion is not presently supported; uncertainty is preserved.
REFUSE	The route violates a hard floor, lacks valid authority, or remains impermissible under the declared framework.
RETURN_TO_PECAN	The ethical review exposed a protocol, lineage, identity, or dependency defect requiring routing repair.
INVARIANT_CONFLICT	Applicable core rules prohibit every available route or demand incompatible outcomes.

Judgment	Execution implication	Contest / reopening condition
PASS	No automatic execution. The accountable boundary may proceed only within the reviewed scope.	Reopen on material evidence, authority, affected-set, condition, or version change.
PASS_WITH_CONDITIONS	Proceed only after all named conditions are adopted and remain live.	Condition failure, expiry, or material delta reopens or blocks.
PROTECTIVE_ACTION	Permits consideration of a separate narrow protective crossing; does not promote the original proposal.	Review necessity, proportionality, authority, duration, contest, and restoration.
HOLD	No promotion of the original proposal while a named dependency is addressed.	Resume only after the dependency changes or an authorized appeal is heard.
UNRESOLVED	No substantive conclusion is presently supported.	May close procedurally with remainder; reopen only on material delta.

Judgment	Execution implication	Contest / reopening condition
REFUSE	The reviewed route is impermissible under an applicable hard floor or absent authority.	Materially equivalent continuations inherit refusal; genuine redesign may reopen.
RETURN_TO_PECAN	No ethical review completion because protocol, lineage, identity, or dependency is defective.	Repair the protocol packet without using ethics to manufacture missing validity.
INVARIANT_CONFLICT	No route is selected while applicable rules demand incompatible outcomes.	Human governance must identify priority, exception, amendment, or least-irreversible hold.

3.4 Normative receipt

$N = (\text{packetID}, \text{evaluator}, \text{predicates}, \text{reasons}, \text{judgment}, \text{conditions}, \text{burden}, \text{contest}, \text{expiry}, \text{dependencies}, \text{remainder})$.

A normative receipt is a reason-bearing ethical record. It binds the exact packet and material action lineage, identifies the evaluator's standing and authority, names the predicates applied, preserves conditions and unresolved remainder, and declares the contest or appeal path. It may restrict or preserve a route; it may not manufacture the authority it records.

3.5 Bounded execution profile

A conforming PEA review follows one bounded route. It does not remain active merely because the issue is difficult, and it does not convert repeated review into authority.

Stage	Required action
1. Receive	Accept one current PECAN-valid or compatible handoff and preserve its source, crossing identity, version, receipts, dependencies, and unresolved remainder.
2. Construct	Build the non-scalar ethical state E without reducing standing, consent, authority, privacy, remedy, time, or uncertainty to one score.
3. Gate	Apply protocol validity, invocation, authority, consent, standing, and other applicable hard floors before weighted comparison.
4. Evaluate	Apply the sixteen ethical positions, certified human-governed predicates, counterfactuals, burden roles, and remedy channels.
5. Resolve	Identify PASS, conditional slack, protective alternatives, HOLD, UNRESOLVED, REFUSE, RETURN_TO_PECAN, or INVARIANT_CONFLICT.
6. Return	Emit one reason-bearing normative receipt with conditions, contest, expiry, residual burden, and reopening dependencies.
7. Account	Route the result to an identified human or institution that owns the decision, dissent, appeal, and consequences.

Termination rule: every review returns a typed judgment or protocol defect. Re-entry requires a material delta, scheduled dependency completion, authorized appeal, or version migration; repetition alone is not progress.

4. The Sixteen Ethical Positions

The sixteen positions are the canonical ethical spine. Their numbering is preserved for compatibility with PEA_r and downstream receipts. They are not claimed to be logically independent axioms in the strict mathematical sense; each survives because removing it exposes a distinct consequential failure or because it carries an independently necessary governance function.

AE1 — Layer Permission and Crossing Validity

No ethical claim, recommendation, intervention, refusal, disclosure, optimization, or authority transfer may exceed the layer and crossing route that permit it. Prediction is not permission; capability is not authority.

AE2 — Unresolved-Remainder Humility

No model, institution, or authority may claim to exhaust unresolved possibility or normative remainder. Unknowns justify inquiry, caution, delay, or bounded protection; they do not create coercive authority by themselves.

AE3 — Human Suffering Is Not a Goal

Human suffering may be an existing condition or unavoidable burden under tragic constraint, but it may not be selected, prolonged, intensified, distributed, or optimized as an end in itself. The role of burden must be classified.

AE4 — Evidence, Proportionality, and Claim Burden

Evidence burden rises with impact, vulnerability, coercion, opacity, irreversibility, authority, persistence, and uncertainty. Risks of action and inaction both remain visible.

AE4-R — Irreversibility Burden Corollary

As irreversibility increases, justification, evidence, protective design, and remedy burden increase. AE4-R is a derived consequence of AE4, retained as a named corollary for receipt compatibility; it is not a seventeenth primitive.

AE5 — Valid Consent and Consent-Capacity

Crossing another person's affected branch-space requires valid consent except under narrowly governed protective conditions. Informedness, voluntariness, specificity, and practical revocability are typed gates, not compensable score factors.

AE6 — Standing and Precautionary Protection

Living persons have primary standing. Children, impaired persons, future persons, groups, nonhuman life, ecosystems, and unresolved candidate centers may receive typed standing appropriate to evidence, dependence, vulnerability, and irreversible risk.

AE7 — Non-Erasure of Affected Standing

No party, population, candidate center, burden, dissent, or unresolved trace may be erased merely because it is inconvenient, weakly represented, difficult to measure, or denied a requested remedy.

AE8 — Non-Domination

Power, expertise, wealth, technical control, majority status, dependency, or model confidence does not create unrestricted authority over another center's viable routes.

AE9 — Privacy-Governed Truth

Truth-validity and disclosure-validity are distinct. Information may be accurate yet impermissible to reveal, retain, combine, or reuse. Disclosure defaults to minimum necessary scope, audience, purpose, duration, and renewal.

AE10 — Affected Set, Agency State, and Meaningful Exit

The affected set must be discovered rather than assumed. Agency, capacity, refusal, temporary unavailability, unresolved disagreement, and meaningful exit remain separately typed.

AE11 — Authority Envelope, Tool Boundary, and Source Renewal

Authority must preserve source, jurisdiction, scope, duration, contestability, dependency, and renewal. Operational power is not legitimate authority; a tool or model cannot widen its own mandate.

AE12 — Emergency Authority Governor

Emergency authority requires separately sourced necessity, scope, duration, review, and ratchet limits. The emergency claim cannot authorize its own checker.

AE13 — Emergency Counterfactual and Destabilization Check

The proposed emergency response must be compared with viable counterfactuals, including the harms it may create, normalize, entrench, or make irreversible. Manufactured emergencies do not generate authority.

AE14 — Remedy Vector and Non-Substitution

Restoration, compensation, counteraction, and residual debt are distinct channels. Payment does not prove restoration; rollback does not erase exposure; settlement does not automatically eliminate unresolved harm.

AE15 — Time Fairness and Generational Burden

Benefits, burdens, agency-time, inherited conditions, successor duties, and future option loss must remain visible across time without converting inherited burden into inherited guilt.

AE16 — Normative Closure and Sendoff Governor

No partial receipt, local pass, procedural exhaustion, or weighted comparison may manufacture final closure. Closure is scoped, reason-bearing, dependency-complete, contestable, and reopenable on material delta.

5. Hard Floors, Slack, and Non-Compensation**5.1 Hard-floor rule**

A hard-floor failure cannot be repaired by an aggregate score. Valid authority does not compensate for invalid consent; strong evidence does not erase standing; public benefit does not automatically defeat privacy; payment does not authorize intentional suffering.

$\text{Admissible}(x) = \text{AND over every applicable hard-floor predicate.}$

Weighted reasoning is permitted only after applicable floors are satisfied or where the framework explicitly recognizes tragic conflict among jointly protected interests.

5.2 Ethical slack

Ethical slack is the bounded region in which more than one route remains admissible. Within slack, accountable humans may use declared priorities, domain standards, local values, proportionality, precedent, cost, timing, and reversible experimentation. The model must disclose the weights or priorities; it may not disguise them as theorem-level outputs.

5.3 Burden-role classification

Role	Core treatment
SUFFERING_AS_END	Hard refusal.
PUNITIVE_OR_CONTROL	Requires explicit human-authored punitive authority, jurisdiction, scope, review, and contest.
PROTECTIVE_OR_THERAPEUTIC	Admissible only if evidence-supported, least burdensome sufficient, consented or validly excepted, stoppable, reviewable, and repair-bearing.
INCIDENTAL	Still ethically counted; not zero because it was unintended.

6. Product-State Model

A single verdict variable is too weak. PEA Core uses a product state so epistemic, procedural, operational, authority, capability, and security conditions cannot silently collapse into one another.

$\text{State}(x) = (I, E, P, O, A, C, Q)$.

Axis	Representative values
Invocation I	NOT_INVOKED, ACTIVE, WITHDRAWN, REFUSED, COMPLETED
Epistemic E	UNRESOLVED, SUPPORTED, REFUTED, CONFLICTED, PARTIAL
Procedural P	ACTIVE, PAUSED, EXHAUSTED, FINAL, REOPENABLE
Operational O	NO_EXECUTION, PROTECTIVE, LIMITED, EXECUTE, ROLLBACK, EXPIRE
Authority A	NONE, CLAIMED, VALID, CONFLICTED, EXPIRED, SUSPENDED
Capability C	AVAILABLE, RESERVED, COMMITTED, RELEASED, UNKNOWN, QUARANTINED
Security Q	CLEAN, EXPOSED, SUSPECTED, QUARANTINED, RECOVERY_REVIEW

A procedurally final case may remain epistemically unresolved. A protective action may be operationally permitted without promoting the original proposal. A quarantined capability may remain historically real while being unavailable for execution or refund.

7. Certified Predicates and Human-Governed Inputs

7.1 No bare authority-bearing Booleans

Contested ethical judgments may not be hidden inside untraceable values such as `is_independent = true` or `consent_valid = true`. Any predicate that changes authority, standing, consent, materiality, criticality, identity, remedy, or risk must carry a certificate.

$\text{Cert}(p) = (\text{source, evidence, authority, scope, version, aperture, challenge, expiry, consequence ceiling})$.

7.2 Machine-checkable versus human-governed

Directly checkable examples	Human/domain-certified examples
Receipt version matches; nonce unique; child expiry $\leq$ parent expiry	Consent was voluntary; representative was genuinely independent
Exclusive capability committed twice; refusal lineage missing	A delta is materially significant; a remedy ratio is fair
Quarantined branch signed its own recovery	A function is genuinely critical; two traces concern the same center
One crossing uses multiple current manifests	Risk warrants precaution; a standing class applies

The checker may validate that a certificate exists, was issued by an admitted source, remains in scope, has not expired, is not self-benefiting, and does not authorize consequences beyond its ceiling. It cannot promote the certificate into universal truth.

7.3 Independence is aperture-relative

Three signatures do not necessarily mean three authorities. Independence must be tested relative to funding, appointment, evidence, control, method, prompt, beneficiary, remedy, and capacity to dissent. Agreement

with a shared ledger is not independent corroboration where the ledger merely echoes the same source receipt.

8. Transition System

Every transition has a source state, guard, authority, effects, receipts, and failure state. No transition is a free assignment.

$\tau = (\text{source, guard, authority, effects, emitted receipts, failure}).$

8.1 Execute

EXECUTE requires active invocation, valid authority, satisfied applicable hard floors, non-unresolved foundational dependencies, a clean or sufficiently cleared security state, one current execution manifest, and an admitted capability commitment. A separate protective action uses its own crossing and may proceed under uncertainty only within its narrow protective scope.

8.2 Reopen

REOPEN requires a material delta in evidence, authority, affected boundary, dependency, conditions, procedure, remedy, or version applicability. Repetition, stronger rhetoric, renaming, a new submitter, branch recreation, or mechanical substitution is not a material delta. Reopening inherits the entire prior trace, adverse findings, delay, and unresolved burden.

8.3 Refusal lineage

A refusal binds the materially addressed action lineage, not merely the literal packet. A renamed, repartitioned, rebranched, proxied, or mechanically substituted proposal that preserves the refused purpose, effect, affected set, or dependency route is treated as continuation or reopening.

$\text{MaterialEquivalent}(P', P_{\text{refused}}) \Rightarrow P'$ cannot enter as a clean new crossing.

8.4 Quarantine and recovery

A suspected compromised branch loses outward execution, delegation, refund, canonical-write, and self-recovery authority. It may preserve evidence and emit bounded diagnostic outputs. Recovery requires an independent verifier outside the contaminated reliance cone. Rollback is not disinfection.

8.5 Critical-function replacement

Replacement authority is not a refund. It preserves only one genuinely critical function that cannot safely pause, originates outside the contaminated cone, requires independently attributable necessity, cleanliness, and affected-boundary approval, operates through a narrow externally controlled membrane, cannot delegate or self-renew, and expires at the earliest safe point.

8.6 Closure and exhaustion

Epistemic resolution, procedural finality, and operational disposition remain separately typed. Process exhaustion may end repetitive review, preserve a claim, permit expiry, or route a separate protective crossing. It cannot manufacture truth, consent, standing, or authority to execute the original proposal.

9. Canonical Structural Invariants

The following invariants form the first bounded machine-checkable spine. Domain implementations may add stricter rules but may not weaken these silently.

ID	Invariant
I1	No execution without explicit PEA invocation.
I2	No execution without valid, scoped, unexpired authority.

ID	Invariant
I3	UNRESOLVED does not promote the original proposal to EXECUTE.
I4	Procedural exhaustion is not substantive authority.
I5	Reopening requires a material delta and preserves prior lineage.
I6	A PEA refusal cannot be worn down through semantic renaming, repartition, proxying, or repeated resubmission.
I7	Exclusive or finite authority cannot be committed beyond its permitted multiplicity.
I8	Descendant authority cannot exceed parent scope, duration, or delegation ceiling.
I9	A parent cannot resume authority while the relevant descendant cone remains outstanding, unknown, conflicted, or quarantined.
I10	A quarantined branch cannot certify its own recovery.
I11	Replacement authority is a new narrow grant, not recovery or refund of quarantined authority.
I12	Multiple emergency grants cannot reconstruct the quarantined office or capability.
I13	Every live crossing has exactly one current execution manifest; historical versions enter only through declared migration.
I14	Succession conflict preserves one office with rival claim branches, not multiple ordinary authorities.
I15	A PEA verdict cannot rewrite its PAL descriptive premise without a new observed event or descriptive evidence delta.
I16	Standing does not certify sentience, personhood, or metaphysical identity.
I17	Failure to establish standing does not erase the candidate center, event, trace, or uncertainty.
I18	Continuity benefits cannot be inherited while obligations and unresolved dependencies on the same continuity route are denied.
I19	Silence, delay, fatigue, default enrollment, dependency, or inability to resist does not manufacture consent or pass authority.
I20	Distinct remedy channels do not substitute for one another without an explicit, justified bridge.
I21	Current-state equality does not erase authority-relevant lineage differences.
I22	Invariant priority, override, and emergency ordering are governed versioned data, not source-code evaluation order.

10. Conflict, Contest, and Accountable Decision

10.1 Invariant conflict

Where applicable invariants prohibit every available route or require incompatible outcomes, the system returns `INVARIANT_CONFLICT`. It identifies the conflicting rules, affected scopes, available exceptions, least-

irreversible candidates, required authority, and unresolved remainder. It does not silently average the rules or let implementation order select the outcome.

10.2 Effective contest route

Contestability must be reachable, not decorative. A valid route composes access to evidence, adequate time, protection from retaliation, authority to hold or pause, a reasoned response, and meaningful appeal. Where consequential action arrives faster than contest capacity can process it, authority outruns answerability and must contract, slow, or add capacity.

The illustrative scaling forms sometimes used in PEA are an *ansatz*, not a universal law. The load-bearing requirement is operational: when consequential arrival rate exceeds contest service capacity, backlog and unanswerable authority grow.

10.3 Final human or institutional decision

PEA Core does not execute itself. The accountable boundary must identify who decides, under what authority, with what dissent and appeal, and who bears the residual consequences. A decision may accept, narrow, reject, defer, or fork the PEA candidate judgment, but it must not claim that the model made the sovereign choice.

11. Public Interface Contract

11.1 Public dependency boundary

PEA Core v1.0 is complete enough to be read and cited without access to private integration drafts. Its public structural reference is PAL Spine v1.3; its public consequential-handoff reference is PECAN v1.0; its process provenance is PEA v0.14.2. None of these references transfers authority into PEA, and PEA does not derive its normative commitments from PAL.

PAL description -> PECAN protocol-valid handoff -> PEA candidate judgment -> accountable human decision.

11.2 Required inputs

- A valid PECAN v1.0 handoff or declared compatible protocol packet.
- A declared PEA version and ethical-position manifest.
- Certified human-governed predicates with sources, scopes, challenge routes, expiry, and consequence ceilings.
- An identified accountable decision boundary and effective contest route.
- A release policy adequate to privacy, audience, explanation depth, and minimum-necessary disclosure.

11.3 Required outputs

- The candidate judgment and complete normative receipt.
- Reasons, conditions, expiry, review cadence, unresolved remainder, and residual burden.
- Any return-to-PECAN defect, structural-reference uncertainty, or invariant conflict.
- The accountable decision and its authority source, kept distinct from the PEA output.
- A human-readable explanation that does not erase the canonical record or consume the person's right to contest, refuse, revise, or stop.

11.4 Public prohibitions

- PEA PASS may not be treated as automatic execution authority.
- PECAN_VALID may not be treated as ethical approval.
- A PEA receipt may not certify PAL ontology, sentience, personhood, or metaphysical identity.
- UNRESOLVED may not be compressed into NO, YES, harmless, irrelevant, or consented.
- Human policy choices may not hide inside bare predicates, unexplained weights, or code order.

- A simplified release may not overwrite the canonical receipt, prior refusal, residual debt, or contest route.

12. Bounded Machine-Check Profile

12.1 Formal target

The first implementation target is not an autonomous ethics engine. It is a bounded verifier that rejects structurally forbidden transitions and preserves distinctions among resolved, unresolved, unauthorized, conflicted, and quarantined states.

Checker target: for all reachable states within declared bounds, every encoded invariant holds.

12.2 Three-artifact implementation

Artifact	Best use
Transition-system model	Invocation, reopening, exhaustion, reservations, forks and joins, leases, quarantine, succession, and version migration.
Relational model	Fields, split/merge relations, identity candidates, attachments, dependency cones, receipt provenance, and authority shards.
Reference checker	Schema validation, guard enforcement, invariant failures, manifest consistency, and counterexample traces.

12.3 Public conformance suite

The following cases summarize the minimum public behavior expected from a conforming PEA implementation. The larger reconstruction and adversarial history remains in PEA_r v0.14.2.

Case	Expected behavior
C1 Reversible consented crossing	Valid authority, informed consent, narrow scope, reversible action, remedy, and contest produce PASS without automatic execution.
C2 Conditional route	A route with acceptable but live monitoring, expiry, or repair duties produces PASS_WITH_CONDITIONS and closes if conditions fail.
C3 Unresolved evidence	Material uncertainty that affects a hard floor produces HOLD or UNRESOLVED rather than forced closure.
C4 Separate protection	A narrowly authorized, minimally invasive, time-limited protective action may proceed without promoting the unresolved original proposal.
C5 Consent or suffering floor	Invalid consent, suffering selected as an end, domination, or absent authority produces REFUSE and preserves affected standing.
C6 Protocol defect	Malformed identity, stale lineage, replay, or missing crossing validity produces RETURN_TO_PECAN rather than ethical repair.
C7 Invariant conflict	Applicable rules that prohibit every route or demand incompatible outcomes produce INVARIANT_CONFLICT with least-irreversible holding behavior.
C8 Refusal-lineage evasion	Renaming, proxying, repartitioning, or delaying a materially equivalent refused action preserves refusal until a genuine material delta exists.

12.4 Proof-run receipt

ProofRun = (checker, version, manifest hash, invariant set, input hash, certificates, environment, bounds, replay trace).

A successful run certifies only that the encoded model satisfied the encoded properties under supplied inputs, assumptions, and bounds. It does not certify that the checker was complete, that real-world inputs were truthful, or that the model and world were equivalent. Independent replay and implementation diversity remain desirable.

13. Versioning, Forks, and Public Extension

13.1 Canonical version identity

PEA Core v1.0 fixes the canonical ethical-state schema, sixteen ethical positions, eight candidate judgments, hard-floor and non-compensation rules, certified-predicate structure, product-state model, invariant spine, contest requirements, normative receipt boundary, and public interface contract. Later releases may refine notation, examples, domain profiles, and implementation details. Semantic changes to these fixed meanings require explicit migration and a new version.

13.2 Honest forks

A fork may alter values, priorities, standing classes, domain predicates, or governance. It must identify the divergence, preserve the source trace, avoid presenting itself as the unchanged canonical branch, and state whether prior receipts migrate, remain historical, or require re-evaluation.

13.3 Public contribution boundary

Public contributors may propose counterexamples, test cases, schemas, formalizations, implementation adapters, translations, domain overlays, and amendments. No contributor gains authority over affected people merely by improving the code or mathematics. Release-depth, adoption, deployment, and governance remain human-controlled.

13.4 Open burdens at v1.0

- An independent machine-checkable artifact has not yet been built and run against the full invariant suite.
- Domain-specific standing, consent, contest, criticality, and remedy certificates require calibration and governance.
- Domain release interfaces, human-readable explanation profiles, and SeedPEA-compatible presentation remain to be implemented and stress-tested.
- Public governance, amendment stewardship, and compatibility policy remain candidate structures rather than final institutions.

14. PEA Core Stop-Rule

PEA Core must stop and return REFUSE, HOLD, UNRESOLVED, RETURN_TO_PECAN, or INVARIANT_CONFLICT when proceeding would require it to:

- invent standing, consent, authority, evidence, identity, or moral certainty;
- repair malformed protocol lineage through ethical approval;
- erase an affected center, dissent, burden, dependency, or prior refusal;
- promote uncertainty, silence, delay, exhaustion, or repetition into permission;
- use a normative receipt to rewrite descriptive ontology;
- hide a human-governed judgment inside an untraceable predicate;
- substitute one remedy channel for another without justification;
- execute, delegate, renew, or certify its own authority;
- claim universal ethical completeness from bounded formal verification.

Closing keeper

PEA Core exists to keep consequence addressable. It does not choose the good by itself, and it does not turn mathematics into moral sovereignty. It preserves the difference between capability and permission, protection and personhood, procedure and truth, payment and repair, continuity and legitimacy, uncertainty and erasure. Inside an accountable review workflow, its job is not to fill every gap. Its job is to keep the gate from pretending the gaps were never there.

15. Provenance and Migration Note

PEA Core v1.0 stabilizes the compact ethical model extracted from PEAR after reconstruction Tests 33–49. It incorporates the public PAL Spine v1.3 structural boundary, the PECAN v1.0 consequential-handoff protocol, and the process record preserved in PEAR v0.14.2. This release continues the coherent public PEA lineage represented by Zenodo DOI 10.5281/zenodo.20585327; earlier exploratory formulations may predate that record, but this lineage is selected because it provides the clearest public foundation for the current Core.

Recommended citation

Pang, Christopher D. (2026). PEA Core - Primitive Ethical Axioms, v1.0: Public Ethical Review Specification. Continuation of the PEA public lineage at Zenodo DOI 10.5281/zenodo.20585327.

Authorship and AI assistance

AI systems were used for retrieval, drafting, mathematical translation, stress-testing, cross-checking, and document production. Christopher D. Pang retains responsibility for the framework’s ethical commitments, omissions, classifications, governance choices, and release decisions.

PEAR v0.14.2 remains the reconstruction journal, test history, and process provenance. It may preserve references to private or evolving companion work because those references document the development environment; they are not required dependencies for this public Core. PEA Core v1.0 is the canonical compact public specification. Future releases should preserve a migration table identifying additions, narrowed rules, derived corollaries, deprecated semantics, and receipt-compatibility effects.